(b) What is the probability that the number of pounds of bread that will be sold tomorrow is

(i) More than 10 pounds,

(ii) Less than 10 pounds, and

(iii) Between 5 and 15 pounds?

Solution:

(a) We know that $\int_{-\infty}^{\infty} f(x)dx = 1$

$$\int_0^{10} Axdx + \int_{10}^{20} A(20-x)dx = 1$$

$$A\left\{\left[\frac{x^2}{2}\right]_0^{10} + \left[20x - \frac{x^2}{2}\right]_{10}^{20}\right\} = 1$$

$$A\left[(50-0)+(400-200)-(200-50)\right] = 1$$

$$A = \frac{1}{100}$$

(b) (i) The probability that the number of pounds of bread that will be sold tomorrow is more than 10 pounds is given by

$$P(10 \le X \le 20) = \int_{10}^{20} \frac{1}{100}(20-x)dx$$

$$= \frac{1}{100}\left[20x - \frac{x^2}{2}\right]_{10}^{20}$$

$$= \frac{1}{100}\left[(400-200)-(200-50)\right]$$

$$= 0.5$$

(ii) The probability that the number of pounds of bread that will be sold tomorrow is less than 10 pounds, is given by

$$P(0 \le X < 10) = \int_0^{10} \frac{1}{100} xdx$$

$$= \frac{1}{100}\left[\frac{x^2}{2}\right]_0^{10}$$

$$= \frac{1}{100}(50-0)$$

$$= 0.5$$

(iii) The probability that the number of pounds of bread that will be sold tomorrow is between 5 and 15 pounds is

$$P(5 \le X \le 15) = \int_5^{10} \frac{1}{100} xdx + \int_{10}^{15} \frac{1}{100}(20-x)dx$$

$$= \frac{1}{100}\left[\frac{x^2}{2}\right]_5^{10} + \frac{1}{100}\left[20x - \frac{x^2}{2}\right]_{10}^{15}$$

$$= 0.75$$

Exercise 6.1

1. Construct cumulative distribution function for the given probability distribution.

X	0	1	2	3
$P(X = x)$	0.3	0.2	0.4	0.1

2. Let X be a discrete random variable with the following p.m.f

$$p(x) = \begin{cases} 0.3 & for\ x = 3 \\ 0.2 & for\ x = 5 \\ 0.3 & for\ x = 8 \\ 0.2 & for\ x = 10 \\ 0 & otherwise \end{cases}$$

Find and plot the c.d.f. of X.

3. The discrete random variable X has the following probability function

$$P(X = x) = \begin{cases} kx & x = 2, 4, 6 \\ k(x-2) & x = 8 \\ 0 & otherwise \end{cases}$$

where k is a constant. Show that $k = \frac{1}{18}$

4. The discrete random variable X has the probability function

X	1	2	3	4
$P(X = x)$	k	$2k$	$3k$	$4k$

Show that $k = 0{\cdot}1$.

5. Two coins are tossed simultaneously. Getting a head is termed as success. Find the probability distribution of the number of successes.

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